

Viscoelastic softening of copper and zinc at high homologous temperatures under seismic-frequency cyclic loading

Tara R. McElhinney^a,*, Andrew M. Walker^b, Martha G. Pamato^c, Oliver T. Lord^d,
Isra S. Ezad^e, Simon A. Hunt^a

^a Department of Materials, University of Manchester, Oxford Road, Manchester, M13 9PL, UK

^b Department of Earth Sciences, University of Oxford, South Park Road, Oxford, OX1 3AN, UK

^c Department of Geosciences, University of Padova, Via Giovanni Gradeno, Padova, 6, 35131, Italy

^d School of Earth Sciences, University of Bristol, Queens Road, Bristol, BS8 1RJ, UK

^e School of Earth and Ocean Sciences, University of Western Australia, Stirling Highway, Crawley, Perth, 6009, Western Australia, Australia

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ABSTRACT

There is an absence of high pressure experimental data relevant to seismology on metals and alloys close to the liquidus, which hampers our understanding of the composition and structure of the Earth's inner core. Here, we address this by performing high-pressure cyclic deformation experiments on hexagonal close-packed zinc and face-centred cubic copper up to melting, as analogues for iron. Samples, together with an elastic reference, were heated to the liquidus at 3–4 GPa while simultaneously being subjected to cyclic deformation at strains of 10^{-4} – 10^{-5} at frequencies of 0.1–0.3 Hz. X-radiographs were used to monitor changes in sample length throughout the experiments, from which E_ω (the finite-frequency Young's modulus) was derived. We report a phenomenon in which both samples exhibit significant softening under approximately constant stress amplitudes at homologous temperatures above ~ 0.85 . Softening is consistent with previous experimental reports of rapid recrystallization in metals. The alignment of E_ω with the minimum elastic Young's modulus suggests that small sinusoidal stresses may have the ability to induce a preferred orientation in metals near the liquidus.

1. Introduction

Seismic waves reveal a complex inner core. P-wave velocity (V_p) varies strongly with depth and direction, with pronounced anisotropy producing substantial directional variations in V_p (Singh et al., 2000). In contrast, S-wave velocity (V_s) is anomalously low relative to that expected for pure iron under inner-core conditions, while the observed V_p can exceed that predicted for pure iron (Dziewonski and Anderson, 1981; Chen et al., 2014; Costa de Lima et al., 2023). The inner core exhibits high attenuation, with estimates of compressional attenuation ranging from 150–300 in the upper inner core and increasing to > 1000 at greater depths (Doornbos, 1974; Li and Cormier, 2002; Cormier and Li, 2002; Romanowicz and Mitchell, 2015). In global 1D models, such as PREM, the inner-core has Q_μ (shear attenuation) of 84 and Q_k (compressional attenuation) of 1328 (Dziewonski and Anderson, 1981), whereas an ideal elastic body has $Q = \infty$.

In principle, core attenuation depends on crystal structure, temperature and the proximity to melting, and larger scale microstructure. Under ambient conditions, body-centred cubic (bcc) iron is stable,

transforming to the face-centred cubic (fcc) structure at high temperatures and the hexagonal close-packed (hcp) structure at high pressures (Yoo et al., 1995). It is widely accepted that the inner core is composed of an hcp-structured iron alloy (e.g. Anderson, 1986; Vočadlo et al., 1999, 2009; Tateno et al., 2012; Sakamaki et al., 2016). However, it has been suggested that, at high pressure, iron may revert to the bcc or fcc structures (e.g. Ross et al., 1990; Vočadlo et al., 1999; Belonoshko et al., 2003; Dubrovinsky et al., 2007; Vočadlo et al., 2008; Côté et al., 2012; Liu et al., 2016; Hrubiak et al., 2018; Zhang et al., 2025; Ginestet et al., 2026). The reality may be that the Fe–Ni–light element alloy comprising the inner core exists as a multi-phase assemblage (Belonoshko et al., 2008; Mattesini et al., 2010; Dominijanni et al., 2022). In part, the uncertainty in the phase of iron in the inner core is due to the extreme pressure and temperature ($P > 300$ GPa and $T > 5000$ K, Ahrens et al., 2002). These conditions are difficult to access experimentally, leading to a lack of data on the behaviour of metals under inner core conditions. Hcp iron is unrecoverable from high pressure and only stable on the

* Corresponding authors.

E-mail addresses: tara.mcelhinney@manchester.ac.uk (T.R. McElhinney), simon.hunt@manchester.ac.uk (S.A. Hunt).

liquidus >100 GPa making performing experiments on relevant phases at realistic conditions challenging.

Melting of metals at high pressures remains a subject of debate, with different experimental diagnostics yielding substantially different melting curves (Parisiades, 2021). Diamond anvil cells (DACs), in which the experiments are often undertaken, allow a diverse range of techniques to infer melting. DACs are often combined with laser heating (LH-DAC), which can produce substantial temperature gradients across the sample, particularly along the laser-heating axis, making the exact melting conditions difficult to determine. Many traditional melt detection techniques have been challenged, such as the laser speckle method (Boehler, 1993) which has been suggested to detect the onset of recrystallization rather than true melting (Anzellini et al., 2013). Due to this strong pre-melting disorder, melting is difficult to identify (Dubrovinsky et al., 2007) which makes measurements near the liquidus challenging.

There is additional uncertainty due to little evidence of high T/T_M microstructure (where T is temperature and T_M is melting temperature). Experimental studies on metals, such as Cu and Zn, show that recrystallization can occur at relatively low homologous temperatures ($> 0.6 T/T_M$) (Chang and Himmel, 1966; Humphreys and Hatherly, 2002; Hrubciak et al., 2017). However, its effects become most pronounced at high T/T_M , where enhanced grain boundary mobility leads to rapid microstructural evolution, with many observations reported near solidus conditions ($T/T_M \geq 0.95$) (Kronberg and Wilson, 1949; Anzellini et al., 2013; Guo et al., 2024).

These complexities and uncertainties hamper our ability to constrain parameters necessary for complex viscoelastic models and often leads to the assumption of an elastic inner core for modelling purposes. Attenuation in the inner core is orders of magnitude greater than expected for pure Fe and is therefore indicative of viscoelastic relaxation (Jackson et al., 2000). A practical approach is to use structural analogues for iron that are stable under experimentally accessible conditions.

Few experimental studies have demonstrated viscoelasticity and high attenuation in metals at seismic frequencies (Jackson, 1994; Jackson et al., 2000; Li and Cormier, 2002; David and Jackson, 2018; Hunt et al., 2024). Aside from David and Jackson (2018), these studies are conducted at homologous temperatures < 0.8 . Most high temperature experiments on metals are at frequencies higher than the ~ 0.1 Hz associated with seismic waves (Jackson et al., 2000).

To investigate viscoelastic behaviour at seismic frequencies and elevated homologous temperatures under experimentally accessible pressures, we performed high-pressure experiments (3–4 GPa) on hcp Zn and fcc Cu as analogues for Fe. Small oscillatory compressive stresses were applied at frequencies of 0.1–0.3 Hz. Strain was measured as temperature was increased beyond the liquidus, and the resulting Young's modulus is reported across a wide homologous temperature range. Heating samples to melting enables more accurate determination of T/T_M , addressing uncertainties in high-pressure melting curves of metals (Parisiades, 2021). Although the P – T conditions are significantly lower than those of the inner core, the reduced pressure allows the use of a larger sample assembly, enabling the application of small strains and the extraction of high-quality mechanical data. These experiments are technically challenging and provide a valuable bridge between ambient-pressure spectroscopy and high-pressure deformation studies.

2. Methodology

Experiments were performed in the D-DIA press on beamline X17B2 at the NSLS (Brookhaven National Laboratory, New York) and on the same apparatus following relocation to beamline APS-6-BM-B at the Argonne National Laboratory (APS, Illinois). In each experiment, the finite-frequency Young's Modulus (E_ω) and dissipation (Q^{-1}) are measured using small sinusoidal strains applied simultaneously to a sample and elastic reference material. The experimental apparatus are the same

as those used previously by Hunt et al. (2024), which themselves are developments of the X-radiographic cyclic loading method originally developed by Li and Weidner (2007) to characterize Q^{-1} . In this study, this method was adapted to measure the E_ω of zinc and copper at temperatures up to the liquidus.

2.1. Experimental procedure

Samples of Zn and Cu consisted of 1 mm diameter cylinders, polished to 1–1.3 mm length with flat, parallel ends. Similarly, the elastic standards of Al_2O_3 (for Zn) and CaF_2 (for Cu) consisted of 1 mm diameter rods polished to ~ 1 mm length. All samples and references were commercial grade with 99% purity, except for the CaF_2 reference which was a high purity $> 99.9\%$ natural sample.

Samples and standards were stacked on top of each other, separated by Pt foils and placed inside a standard cell assembly (used previously by Dobson et al., 2012; Hunt et al., 2024). The assembly consists of a 6.1 mm pyrophyllite cube, pre-baked to 1000 °C. Components were inserted into a 3 mm diameter hole within the cube, including an alumina sleeve (ID = 2.36 mm, OD = 3.0 mm, where ID and OD are inner and outer diameters respectively), a graphite furnace (ID = 2.1 mm, OD = 2.36 mm, length = 6.1 mm), and a boron nitride sleeve (OD = 1.8 mm, ID = 1.0 mm, length = 3.0 mm). A C-type thermocouple was inserted through the furnace but was not in contact with the sample itself.

Samples were compressed to the desired pressure and then the assembly was strained sinusoidally at the chosen period with the smallest amplitude that could clearly be resolved. The zinc experiment with an alumina Al_2O_3 standard ($E \approx 340$ GPa) was performed with a 30 s period and heated from 150 °C to ≈ 570 °C for 215 mins. The copper experiment with a CaF_2 standard ($E \approx 75.8$ GPa) was performed with a 10 s period and heated from 25 °C to ≈ 1540 °C over a period of 152 mins. The zinc experiment was heated along a continuous temperature ramp; technical issues meant the heating ramp of the copper experiment was paused around 1250–1300 K for ~ 2 h; during this time the sinusoidal deformation remained running to maintain the sample microstructure and prevent (additional) grain-growth. During heating, X-radiographs were collected using an yttrium aluminium garnet scintillator and visible-light camera, at a rate of 1.5 s/frame for the zinc experiment and 0.5 s/frame for the copper experiment. X-ray diffraction data was collected intermittently during heating of the zinc experiment but not during the copper experiment.

The experiments were quenched when the samples had fully melted, at $\approx 1.1 T/T_M$. There are a number of different criteria for identifying the onset of melting in experiments found in the literature, typically each with its own distinct high pressure melt curve (e.g. Jephcoat and Besedin, 1996; Li and Weidner, 2014; Aquilanti et al., 2015; Stutzmann et al., 2015; Anzellini et al., 2013). In this study melting was identified from changes to sample morphology observed in the X-radiographs. Melting of the Cu sample was identified by a significant change in its shape, from cylindrical to orbicular (Fig. 1b). The zinc sample did not change shape as distinctly because the melting stirred platinum from the foils into the sample forming a Zn-Pt alloy. This alloying increased the melting temperature (see Zn-Pt phase diagram in Moser, 1991), and re-solidified the sample. The stirring of the platinum into the sample is observed in the radiographs as a plume of more absorbing platinum-rich material rising through the sample, as well as fattening of the Pt foils at the onset of melting (Fig. 1a). The different melting morphologies are also attributable to the different standards, the Al_2O_3 standard for Zn experiments is far from its melting point while for the Cu experiment, the CaF_2 standard is at a higher homologous temperature and as such is softer and more easily deformable.

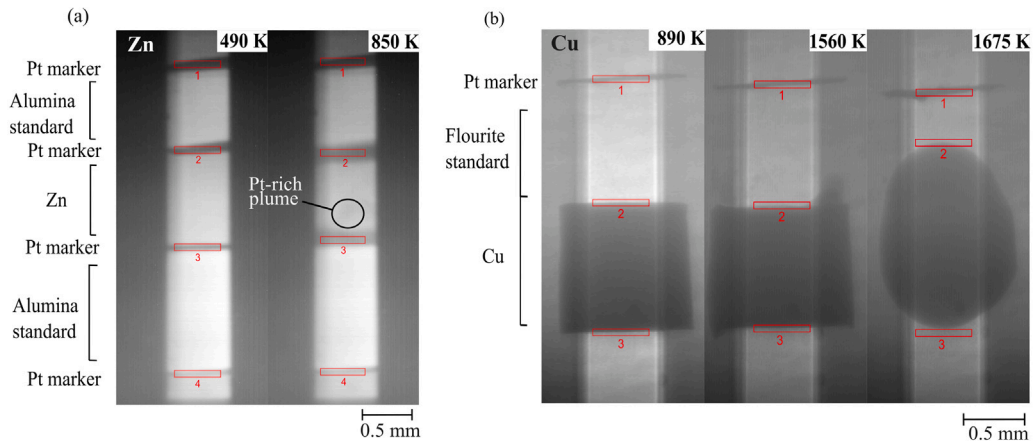


Fig. 1. X-radiograph images of experiments, each sample is ~ 1 mm in length in the first image. The red boxes represent highlighted regions of interest for data processing. (a) Zn experiment before melting at 490 K and after melting at 850 K. The dark patch above the box 3 at 850 K is a plume of molten Pt in the Zn sample, and (b) Cu experiment before melting at 890 K, at the onset of melting at 1560 K, and fully molten at 1675 K.

2.1.1. Pressure and homologous temperature determination

Pressures were determined intermittently during the Zn experiment from energy dispersive X-ray diffraction patterns collected on a 10-element detector that was designed to measure differential stress during high pressure experiments (Weidner et al., 2010). Pressures were determined from the Al_2O_3 reference rather than the metals themselves because the metals recrystallized after heating and the diffraction patterns did not contain sufficient peaks to accurately determine sample volume as per Hunt et al. (2024). Diffraction peaks were fitted using *Plot85* and independent unit cell volumes calculated for each of the detector elements. Volume strains were calculated independently for each of the detector elements using the corresponding open-press unit cell volume, the corundum thermal expansion coefficients of Fei (1995) and the temperature reported by the thermocouple.

Pressure was calculated from the mean Al_2O_3 volume strain, assuming the bulk modulus of $K_0 = 254.28$ GPa along with pressure and temperature derivatives of $K' = (\partial K / \partial P) = 4.27$ and $\partial K / \partial T = -0.0173$ GPaK $^{-1}$ respectively. The bulk modulus and the temperature derivative are a linear fit to the Voigt–Reuss–Hill bulk moduli calculated using *MSAT* (the Matlab Seismic Anisotropy toolbox; Walker and Wookey, 2012) from the elastic stiffnesses (c_{ij}) of Goto et al. (1989). The pressure derivative was calculated from the pressure dependencies of the elastic stiffnesses of Gieske and Barsch (1968) in the same manner, assuming the derivatives are linear at pressures greater than 1 GPa. Elastic stiffnesses were used, rather than an Equation of State, for internal consistency with subsequent E calculations. No observable differential stress effects were seen in the Al_2O_3 diffraction data.

During the Cu experiment, technical difficulties prevented switching between diffraction and imaging modes and so only X-radiographic images were collected and pressures could not be directly determined but were instead inferred from observed melting temperatures. In both cases T/T_M was calculated using the T_M observed, assuming the experiment was isobaric. At low temperatures, absolute values of T/T_M may therefore be incorrect by a few % but this reduces to 0 at melting temperature. Importantly, the relative T/T_M errors between adjacent observations is minimal.

2.2. Post-processing of X-radiographic images

X-radiographs capture the changing position of the marker foils as the experiment is strained and heated. X-radiographs were post-processed to determine the lengths of the sample and reference based on the relative displacement of the Pt foils. Data processing was carried out using *FoilTrack* (Hunt, 2023) in *MATLAB 2021b*. The basis of the algorithm is finding the minimum of the Sum Squared Differences

(SSD) in pixel intensity between highlighted regions of interest (Fig. 1) in a reference image and comparison image (e.g. Li and Richards, 2003). The improved algorithm in *FoilTrack* treats X-radiographs as a continuous data series, rather than a pair of images in isolation, and the SSD for a sequence as a single polynomial surface. The result is improved precision in measured displacements (Hunt, 2023). Here we process the data in significantly shorter blocks than in previous studies using a moving window of 181 frames, equivalent to 9 sinusoidal cycles in each experiment. This gives the best temperature resolution while minimizing the scatter in the calculated values. This scatter arises primarily from the effect of random and fixed pattern noise in the images (Janesick, 2007; Hopkinson et al., 2004). Attempts to remove the fixed pattern noise from the images did not reduce scatter in the fit and derived values. Consequently, the uncertainty in the derived parameters is not limited by the formal fit errors, but by the intrinsic noise in the imaging system, which sets the effective resolution of the measurements. The fixed pattern noise prevented the measurement of Q.

The *FoilTrack* algorithm returns the amplitude (A), phase and period of the sinusoidal length change in both the sample and the elastic standard, as well as the length (l) of the samples in the reference image. Sinusoidal strain (ϵ), in each window, is calculated from

$$\epsilon = \frac{A}{l} \quad (1)$$

(Li and Weidner, 2007). The uncertainty in foil displacement is governed by image noise and is estimated from the scatter in fitted amplitudes to be on the order of $\sim 10^{-2}$ – 10^{-3} pixels. The amplitude of the stress (σ) applied during sinusoidal deformation is derived from

$$\sigma = \epsilon_{\text{reference}} \times E_{\text{reference}} \quad (2)$$

assuming the elastic standard is behaving elastically, where $E_{\text{reference}}$ is the Young's modulus of the elastic reference at the temperature and pressure of the observation. Values were calculated using *MSAT* (Walker and Wookey, 2012) from the elastic constants of the references for Al_2O_3 (Goto et al., 1989) and CaF_2 (Catlow et al., 1978) respectively for each point. The formal errors associated with the applied stress produced from *FoilTrack* are in the region of 0.04 MPa in Zn and 0.2 MPa in Cu. The errors in stresses are dominated by errors in $\epsilon_{\text{reference}}$. These errors are larger than those in elastic stiffnesses used to calculate $E_{\text{reference}}$, which are not published in all manuscripts, but are on the order of a few percent for Brillouin spectroscopy (Speziale et al., 2008). Typical precision errors in $E_{\text{reference}}$ propagate to identical magnitude errors in $\epsilon_{\text{reference}}$.

This approach relies on the assumption that the sample and elastic reference are mechanically coupled in series and therefore experience

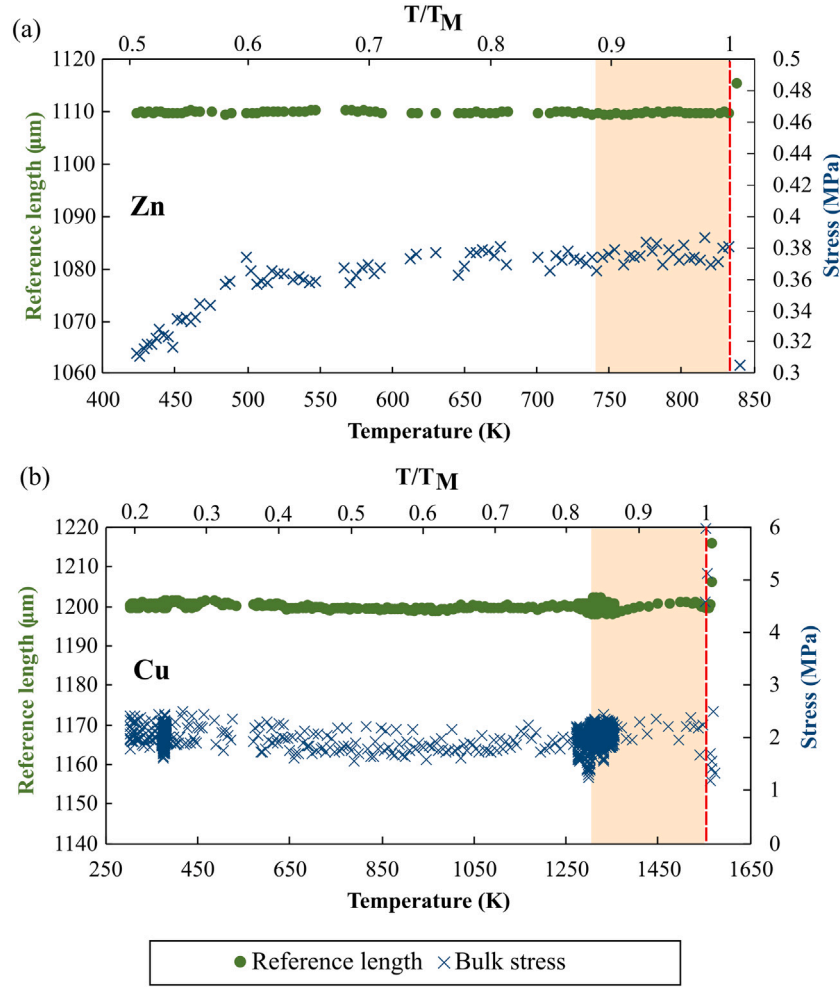


Fig. 2. Length of elastic reference (Eq. (3)) and the resulting bulk stress, plotted against temperature and homologous temperature for Zn and Cu, shaded is the area where softening occurs above $\sim 0.85 T/T_M$. The elastic reference for each experiment undergoes no lengthening or shortening under the approximately constant applied stress. Stress increases slightly for the first 150–200 K of the Zn experiment but then remains at 0.38 ± 0.1 MPa for the remainder of the heating ramp.

the same axial stress during cyclic deformation. Strain is partitioned between the sample and reference according to their relative stiffnesses. We further assume that (i) no significant stress gradients develop along the length of the assembly, (ii) interfacial slip between the sample and reference is negligible, and (iii) the elastic reference remains within its linear elastic regime over the full range of applied sinusoidal loading.

The finite frequency Young's modulus, E_ω , was then derived from

$$E_\omega = \frac{\epsilon_{\text{reference}}}{\epsilon_{\text{sample}}} E_{\text{reference}} \quad (3)$$

The E_ω values calculated here are compared to the elastic (infinite frequency) (E_∞) measurements. The isotropic polycrystalline average (E_∞^{iso}), the minimum single crystal (E_∞^{min}), and maximum single crystal (E_∞^{max}) possible E of the samples were determined from c_{ij} s using MSAT (Walker and Wookey, 2012). For Zn, the ambient condition elastic stiffnesses and their temperature dependencies were taken from Alers and Neighbours (1958). For Cu, the ambient condition c_{ij} s were extracted from Daniels and Smith (1958) while the temperature derivatives were taken from Ledbetter and Naimon (1974).

Uncertainty in pressure has only a limited effect on the inferred stress amplitudes. The applied stress is calculated from the strain in the elastic reference and its temperature-dependent Young's modulus (Eq. (2)), and therefore does not explicitly depend on pressure. While pressure may influence the elastic properties of the reference material,

the pressure dependence of the elastic constants in metals is comparatively weak in this modest pressure regime, with experimentally measured derivatives $\delta c_{ij}/\delta P$ corresponding to only a few percent change per GPa in Cu and Zn (e.g. Daniels and Smith, 1958; Swartz and Elbaum, 1970). Furthermore, these pressure derivatives decrease with increasing temperature (Brugger, 1964), over the 3–4 GPa range investigated here, pressure-induced variations in c_{ij} s are expected to be small compared to thermal effects. For the Cu experiment, where pressure is inferred from the melting curve, the estimated uncertainty of 0.6 GPa therefore contributes negligibly to uncertainty in stress amplitude.

3. Results

We report here the first experimental measurements of E_ω up to the liquidus in metals at high pressure. In zinc, melting was observed at 565 °C (838 K), as indicated by mixing of platinum into the sample (Fig. 1a). From the zinc melting curve ($T_{M,0} = 419.5$ °C, $dT_M/dP = 40$ K/GPa Errandonea, 2010), this temperature corresponds to 3.6 GPa. During heating, the pressure of the Zn sample was measured by diffraction from corundum at temperatures of 27 °C, 225 °C, 268 °C, 289 °C, and 422 °C. Assuming the temperature in the corundum matched the thermocouple readings, the corresponding pressures were 3.8(5), 3.5(3), 3.5(2), and 3.4(3) GPa. A linear fit of pressure against temperature gives a pressure reduction of 1.04 MPa/K, which intersects the melting curve

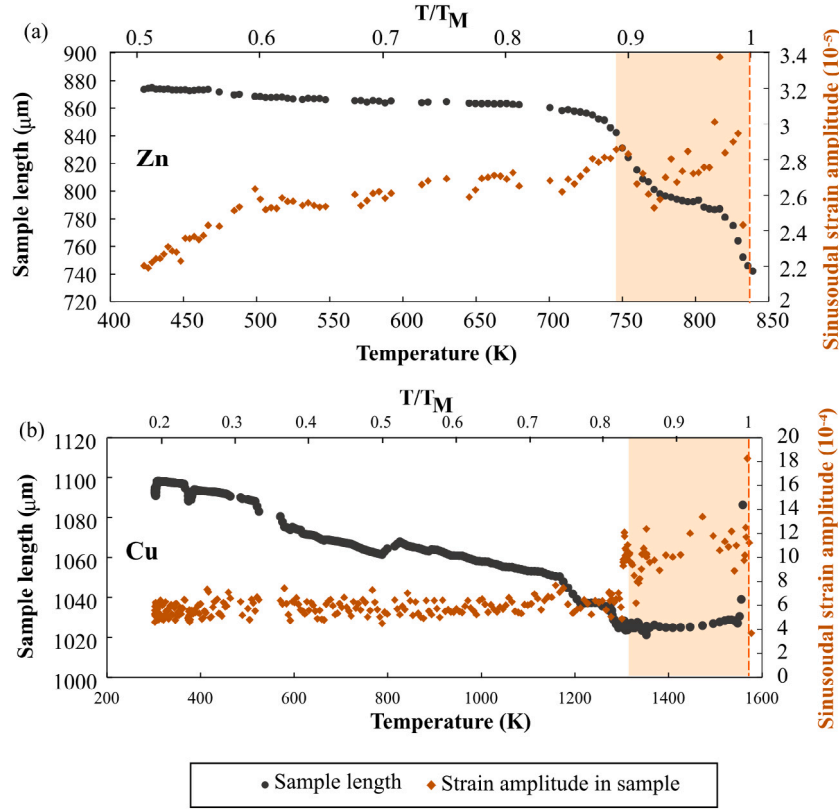


Fig. 3. Plot of strain amplitude in the sample and sample length (Eq. (1)) as a function of temperature in (a) Zn and (b) Cu experiments. The shaded area in orange represents the softened sample.

at 550(13)°C (823(13) K) and 3.4(3) GPa (Errandonea, 2010). Copper melted at 1278 °C (1551 K), as shown by sample changing shape in Fig. 1b. This T_M corresponds to a pressure of 4.6 ± 0.6 GPa, using the (experimental resistance) melting curves and errors of Brand (2005) and Errandonea (2010).

The applied sinusoidal deformation did not change throughout the tests. This is reflected in the amplitude of the stress (Eq. (3)) in both the Zn and Cu samples, which remains approximately constant. In the zinc experiment, the applied stresses are 0.32–0.38 MPa and strains are $\sim 10^{-5}$. After an initial increase above $T/T_M = 0.6$, stresses remain at 0.38 ± 0.01 MPa until the liquidus (Fig. 2a). In copper, bulk stresses are 2–3 MPa remaining approximately constant throughout the experiment, these stresses correspond to strains of $\sim 10^{-4}$ (Fig. 2b). While the stresses applied here correspond to strains larger than the 10^{-6} – 10^{-7} imparted by seismic waves (Jackson et al., 1992), and larger than the $\sim 10^{-5}$ needed for viscoelastic consistency (Hunt et al., 2024). They are the smallest strains that could be resolved in these experiments.

Despite the approximately constant bulk stress applied to the cells, the strain in both samples increases between $0.8 < T/T_M < 0.9$ and remains high until the melting point. In both copper and zinc, the increase in strain (Eq. (1)) is caused by a decrease in sample length and a corresponding increase in the amplitude of deformation (Fig. 3). The zinc sample decreases gradually in length until it shortens by 80–90 μm at $T/T_M \sim 0.9$, with an initial decrease in strain amplitude followed by a large increase associated with shortening (Fig. 3a). The copper sample undergoes intermittent shortening and lengthening of ± 10 μm throughout, attributed to pauses in heating during the temperature ramp, and above 0.82 T/T_M the length of the Cu sample decreases by 15–20 μm (Fig. 3b).

The derived E_ω values are plotted in Fig. 4. The E_ω values are calculated directly (Eq. (2)) from the strain amplitudes shown in Fig. 3, such that increases in sample strain correspond to reductions in modulus and define the observed softening trends. The evolution of E_ω

in both Cu and Zn does not simply parallel the elastic Young's modulus trends but shows a broadly consistent pattern between the samples. In both samples, below $T/T_M \sim 0.85$, E_ω follows a decreasing trend approximately parallel to E_∞ . In copper, the data is approximately coincident with the isotropic Young's modulus, whereas in zinc the values are lower, likely due to inherited texture. In both samples, the onset of softening occurs at homologous temperatures of ~ 0.84 – 0.88 , well below the liquidus, demonstrating that significant mechanical weakening precedes melting. In Zn, softening occurs at ~ 752 K (~ 0.88 T/T_M), with E_ω dropping to approximately E_∞^{min} and remaining within ± 10 GPa of this value until melting at ~ 838 K (Fig. 4a). Similarly, in Cu, E_ω values approximately follow the trend of E_∞^{iso} , remaining within 20 GPa until ~ 1280 K (~ 0.84 T/T_M). Above this temperature, Cu softens, and E_ω aligns with E_∞^{min} , until melting at ~ 1558 K (Fig. 4b). At the liquidus, E_ω exhibits significant excursions as the samples change markedly, coincident with melting in X-radiographs. Although the E_ω in any small temperature range shows relatively large scatter (± 8 GPa for Zn, ± 20 GPa for Cu) the change of trend at $T/T_M \sim 0.85$ is larger than those deviations. Therefore we think the signal, measured independently in 2 samples, is robust.

Average errors on E_ω are generally in the range of 5 GPa for Zn and 11 GPa for Cu and are constant throughout the experiments. The formal error is less than the scatter on the data (Fig. 4). We attribute small variations in all measurements throughout the course of the experiment to the effect of fixed pattern noise (Hunt, 2023) in the X-radiographs. It is not appropriate to smooth the E_ω values using a moving average, as neighbouring values are not independent; the data from which they are derived differ by only one radiograph.

4. Discussion

In this study we report measurements of the finite-frequency Young's modulus (E_ω) up to the liquidus in zinc and copper at seismic frequencies (Fig. 4). Unlike elastic moduli derived from static or high-frequency

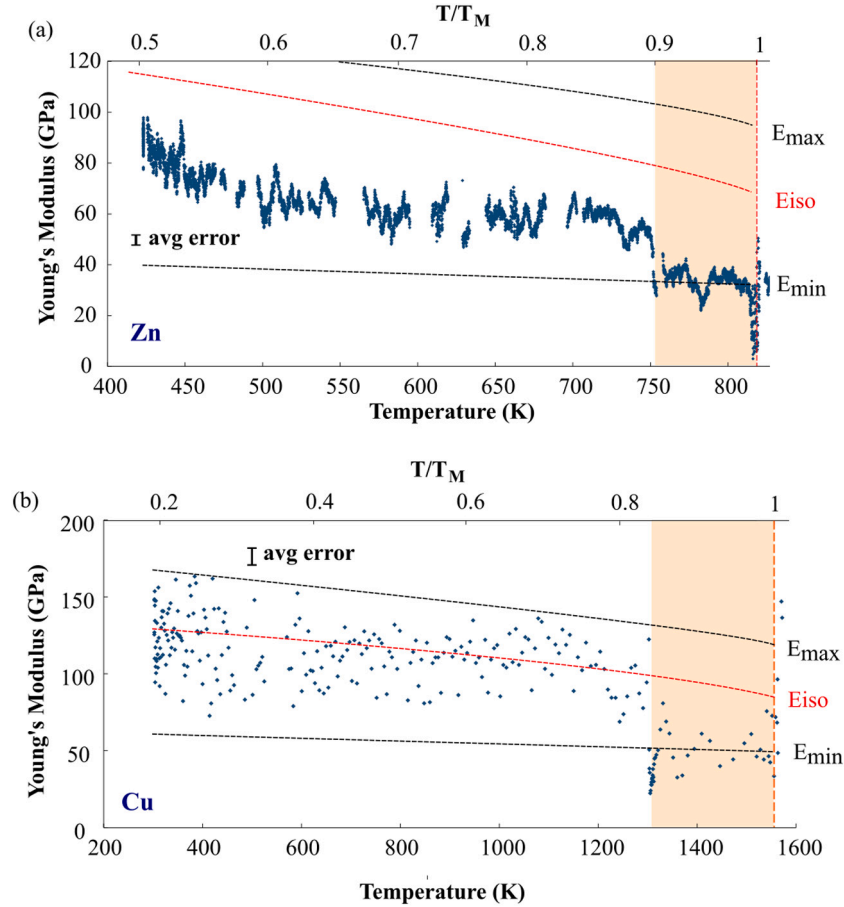


Fig. 4. E_{ω} plotted against temperature and T/T_M for Zn and Cu, results from this study are plotted against the elastic E_{∞}^{iso} , $E_{\infty}^{\min}$ and $E_{\infty}^{\max}$, highlighted in orange is the softening period. There are no errors included in the plot due to the data density, average errors are included in the range of ~ 5 GPa for Zn and ~ 11 GPa for Cu.

measurements, E_{ω} captures the mechanical response of the material under cyclic loading and therefore incorporates both elastic and time-dependent (viscoelastic) deformation. The measured values of E_{ω} lie between the maximum and minimum possible elastic bounds. Below $T/T_M \sim 0.85$, E_{ω} tracks the temperature dependence of E_{∞}^{iso} . Above $T/T_M \sim 0.85$, both samples soften and closely follow $E_{\infty}^{\min}$. In both cases, the reduction in E_{ω} arises from an increase in the amplitude of sample strain under an approximately constant applied sinusoidal stress (Fig. 2). This increase in strain is coincident with sample shortening (Fig. 3), the mechanical change in the sample produces viscoelastic softening.

The onset of softening occurs well below the liquidus and coincides with the homologous temperature range over which dynamic recrystallization (DRX) is expected to become active, the deformation mechanism maps of Frost and Ashby (1982) predict a transition to DRX in many metals at $T/T_M > 0.8$. While recrystallization itself is well established, the results presented here quantify its effect on the finite-frequency response. Changes in grain size alone cannot explain the magnitude and abruptness of the observed softening, nor the convergence of E_{ω} toward the minimum single-crystal modulus. Instead, the alignment of E_{ω} with $E_{\infty}^{\min}$ suggests that deformation is accompanied by the development of a crystallographic preferred orientation, such that the mechanical response becomes dominated by the softest elastic direction.

Experimental studies have also reported recrystallization and associated softening at homologous temperature ranges consistent with the softening reported here (Lord et al., 2014; Zhang et al., 2020; Sinmyo et al., 2019; Hrubciak et al., 2017). Pre-melting softening, reported at $T/T_M > 0.96$ in Fe (Martorell et al., 2013; Zhang et al., 2023) is an intracrystalline phenomenon, and likely represents a separate mechanism of

softening than that reported here which appears to be defect-mediated. This higher T/T_M softening is expected to act cumulatively with the processes identified here, possibly resulting in very soft E_{ω} values near the liquidus. Recrystallization itself is inherently dissipative and can be associated with a finite quality factor (Q), analogous to the behaviour described by Hunt et al. (2024). Unfortunately in the present experiments, the short window length, combined with the influence of fixed pattern noise, prevents a robust determination of Q .

Similar viscoelastic softening has not been universally observed in high-temperature metal experiments. For example, David and Jackson (2018) reported no comparable reduction in Young's modulus. However, their samples were annealed close to melting prior to deformation, resulting in larger grain sizes, and the applied shear strains were significantly smaller than the uniaxial strains used here. This contrast highlights the sensitivity of the viscoelastic response to microstructure, and potentially strain amplitude and deformation geometry. The absence of softening in David and Jackson (2018) suggests that the activation of time-dependent processes is strongly dependent on experimental conditions.

The results demonstrate that processes such as recrystallization, while well known, have a measurable impact on 'elastic' properties when assessed at finite frequency. This provides a direct link between microstructural evolution and seismic-frequency mechanical behaviour, providing a route to the quantification of the effects of high-temperature deformation processes at seismic wave speeds. While the experiments are conducted at pressures significantly lower than those of the inner core and at strain amplitudes approximately two orders of

magnitude larger than seismic strains, they nevertheless provide a qualitative framework for understanding how time-dependent deformation may influence wave propagation in planetary interiors.

5. Conclusion

These measurements provide direct experimental constraints on the finite-frequency elastic response (E_{ω}), linking microstructural processes such as recrystallization to reductions in elastic stiffness and potentially increased dissipation. In a viscoelastic or anelastic medium, seismic wave speeds are lower than those predicted by purely elastic models, and the behaviour observed here provides a mechanism by which such reductions may occur. Softening is observed in both hcp Zn and fcc Cu, indicating that the behaviour is not specific to a single metal or structure, but is characteristic of close-packed metals at high homologous temperatures. The results demonstrate that, at seismic frequencies, metals can exhibit significant reductions in effective elastic modulus due to the activation of time-dependent deformation processes. While extrapolation to Earth's inner core must be treated cautiously, given the lower pressures and higher strain amplitudes of these experiments, the results suggest that similar processes could contribute to reduced seismic velocities and enhanced attenuation under core conditions. If time-dependent deformation processes such as recrystallization operate on seismic timescales, and at the extreme conditions of planetary interiors, they may provide a viable mechanism for viscoelastic relaxation in the inner core.

CRediT authorship contribution statement

Tara R. McElhinney: Writing – original draft, Investigation, Formal analysis. **Andrew M. Walker:** Writing – review & editing, Methodology, Investigation. **Martha G. Pamato:** Writing – review & editing, Methodology, Investigation. **Oliver T. Lord:** Writing – review & editing, Methodology, Investigation. **Isra S. Ezad:** Methodology, Investigation. **Simon A. Hunt:** Writing – original draft, Supervision, Methodology, Investigation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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